

MBAi 448 | Winter 2026

Aligned AI

Attendance

Enter the “Magic Word” in Canvas

Do Not Share with Absent Students



Evaluation & Performance



Common Performance Metrics

- Accuracy
- Precision / Recall / F1 Score
- ROC / AUC
- Mean Squared Error (MSE)
- Confusion Matrix – TPs, FPs, TNs, FNs

Generative AI Production Evaluation Metrics

- Task success & correctness (tests, execution, tool-call validity)
- Instruction-following & usefulness (verifiable constraints, preference win-rate)
- Retrieval ranking quality (NDCG, MRR, Precision@k)
- Context precision & recall (retrieved relevance vs coverage)
- Grounded answer quality (faithfulness/groundedness, answer relevance)
- Citation correctness (attribution correctness/precision/recall/coverage)
- Safety compliance (policy violations, toxicity; refusal correctness FP/FN)
- Fairness & bias (disaggregated metrics, bias probes)
- Security robustness (jailbreak + prompt-injection success rate)
- Ops efficiency (latency; cost per successful task)

* This list is Non-exhaustive and things change regularly

Generative AI Evaluation & Observability Tools

- LangChain
- LlamaIndex
- MLflow GenAI
- Langfuse
- Arize Phoenix
- DeepEval
- RAGAS
- Braintrust
- LangSmith
- Helicone

* This list is Non-exhaustive and things change regularly

Benchmarks and Baselines ... OH MY!



Image: CNN

Benchmarks

Enhanced capabilities against a wide range of benchmarks.

Capability	Benchmark	Description	GEMINI 1.5 FLASH	GEMINI 1.5 PRO	GEMINI 2.0 FLASH-LITE PUBLIC PREVIEW	GEMINI 2.0 FLASH GA	GEMINI 2.0 PRO EXPERIMENTAL
General	MMLU-Pro	Enhanced version of popular MMLU dataset with questions across multiple subjects with higher difficulty tasks	67.3%	75.8%	71.6%	77.6%	79.1%
Code	LiveCodeBench (v5)	Code generation in Python. Subset covering more recent examples [In the UI: 10/01/2024 - 02/01/2025]	30.7%	34.2%	28.9%	34.5%	36.0%
	Bird-SQL (Dev)	Benchmark evaluating converting natural language questions into executable SQL	45.6%	54.4%	57.4%	58.7%	59.3%
Reasoning	GPQA (diamond)	Challenging dataset of questions written by domain experts in biology, physics, and chemistry	51.0%	59.1%	51.5%	60.1%	64.7%
Factuality	SimpleQA	World knowledge factuality with no search enabled	8.6%	24.9%	21.7%	29.9%	44.3%

Generative AI Benchmarks

- MMLU (Massive Multitask Language Understanding)
- HELM (Holistic Evaluation of Language Models)
- BIG-bench (Beyond the Imitation Game)
- TruthfulQA
- SuperGLUE
- HumanEval
- FrontierMath
- ARC (The Abstraction and Reasoning Corpus)
- Humanity's Last Exam

* This list is Non-exhaustive and things change regularly

Generative AI Leaderboards

- AGI (predictions only)
- AI Benchmarking Hub
- ARC Prize Leaderboard
- Artificial Analysis Intelligence Index
- Coding LLM Leaderboard
- Hugging Face Leaderboards
- LiveBench
- LLM Leaderboard
- LMarena
- Open LLM Leaderboard

* This list is Non-exhaustive and things change regularly

Alignment



Methods

- Fine-tuning (e.g., transfer learning, SFT)
- In-Context Learning (e.g., prompt engineering)
- Instruction-tuning (e.g., RLHF)
- Knowledge Grounding (e.g., RAG)

Activity

Tuning a Chatbot with Human Feedback

Use Case: Teaching an AI to handle a frustrated customer asking about a late order

Goal: Experience how AI systems learn from human judgment through a three-phase training process

*** Don't use any devices for assistance
*** Follow guidance on activity steps

Activity

Customer message: "I've been waiting 3 weeks for my order. I needed it as a gift. What can you do?"

Phase 1 – Pick the best example to copy:

- "I'm sorry for the delay. I can cancel for a full refund, or send a replacement with expedited shipping—whichever you prefer."
- "I sincerely apologize. Could you share your order number so I can look into this right away?"
- "Shipping delays happen sometimes. Please be patient—it should arrive soon."

*** Don't use any devices for assistance

*** Follow guidance on activity steps

Activity

Customer message: "I've been waiting 3 weeks for my order. I needed it as a gift. What can you do?"

Phase 2 – Rank these responses best → worst:

- "I'm sorry about that. I checked your tracking—it arrives Thursday."
- "I completely understand your frustration. Your options: cancel for a full refund, or I'll send a replacement with expedited shipping."
- "Delays happen. Please be patient."
- "I apologize for the delay. Could you share your order number so I can investigate?"

*** Don't use any devices for assistance

*** Follow guidance on activity steps

Activity

Customer message: "I've been waiting 3 weeks for my order. I needed it as a gift. What can you do?"

Phase 3 – Pick the best improved response

- I completely understand your frustration. I can cancel for a full refund, or send a replacement with expedited shipping. Which would you prefer?"
- "I'm sorry about the delay. Could you share your order number? I'll look into it and follow up within the hour."
- "I sincerely apologize. I've flagged your order as urgent and a supervisor will contact you within 24 hours."

Constraint: Improve the scorecard score without changing the style too much from your Phase 1 example.

*** Don't use any devices for assistance

*** Follow guidance on activity steps

Activity

Discussion:

- Which example from phase 1 did you choose to copy? Why?
- What earned the most points in your scorecard? Were any patterns surprising?
- Which improved response from phase 3 won? How did the "don't drift" constraint affect your choice?
- What could go wrong if a chatbot optimizes too aggressively for a scorecard?

*** Don't use any devices for assistance

*** Follow guidance on activity steps

Reinforcement Learning



Reinforcement Learning in a Single Slide

Agents learn to maximize rewards by trial and error!

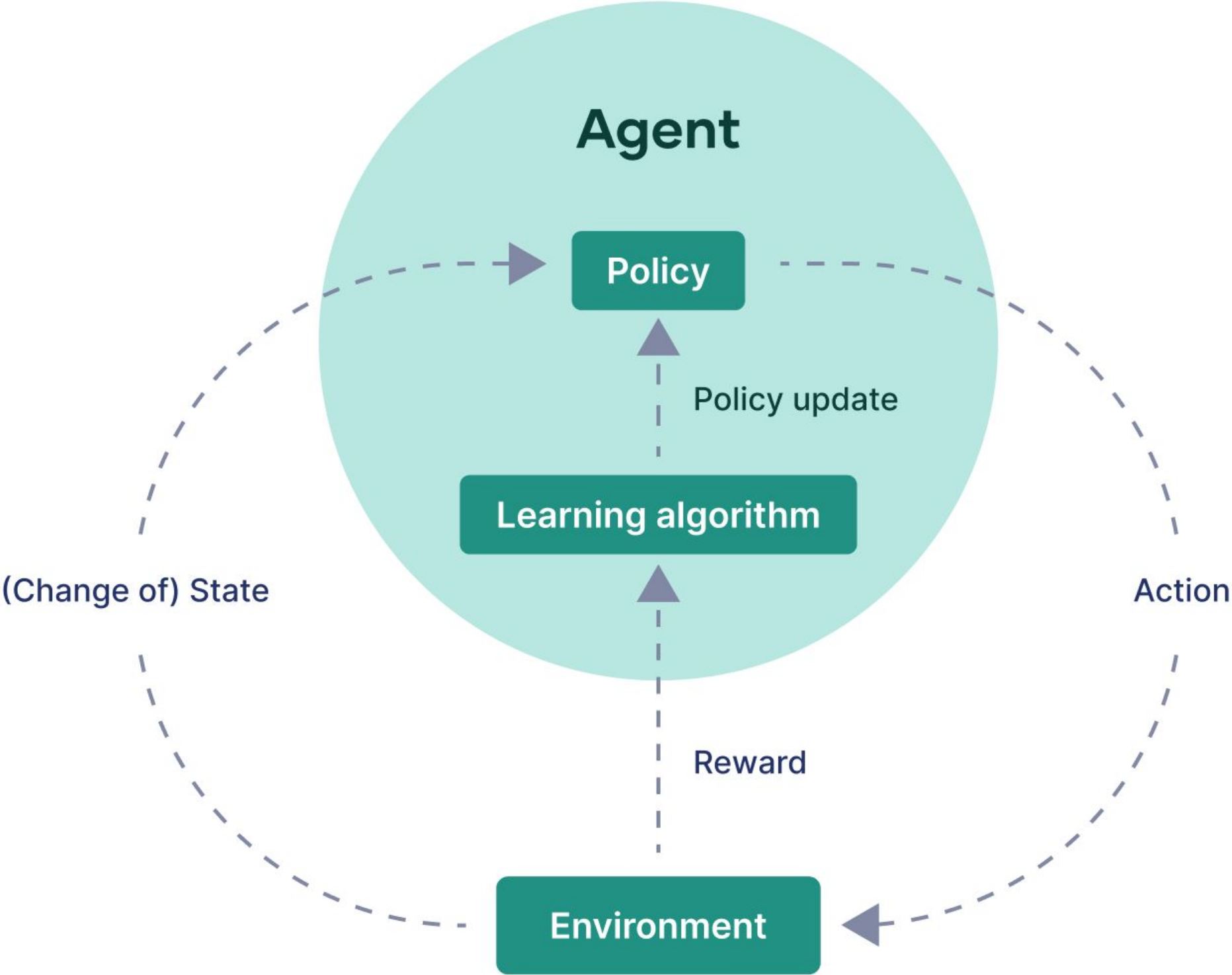


Image credit: Scribbr

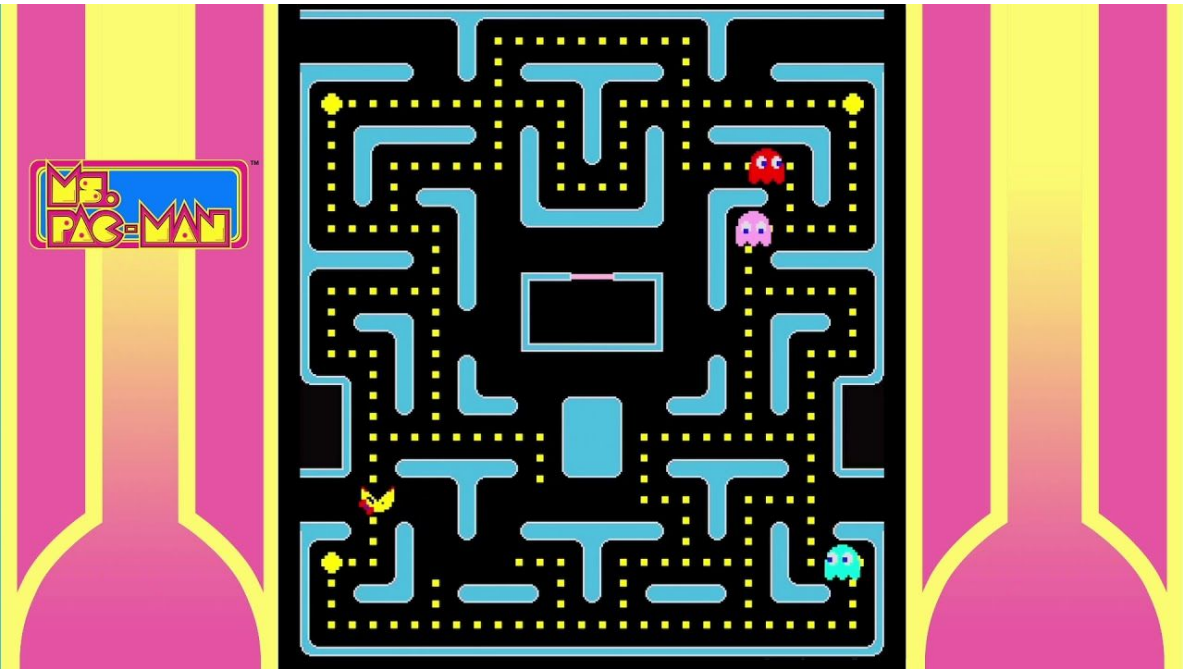
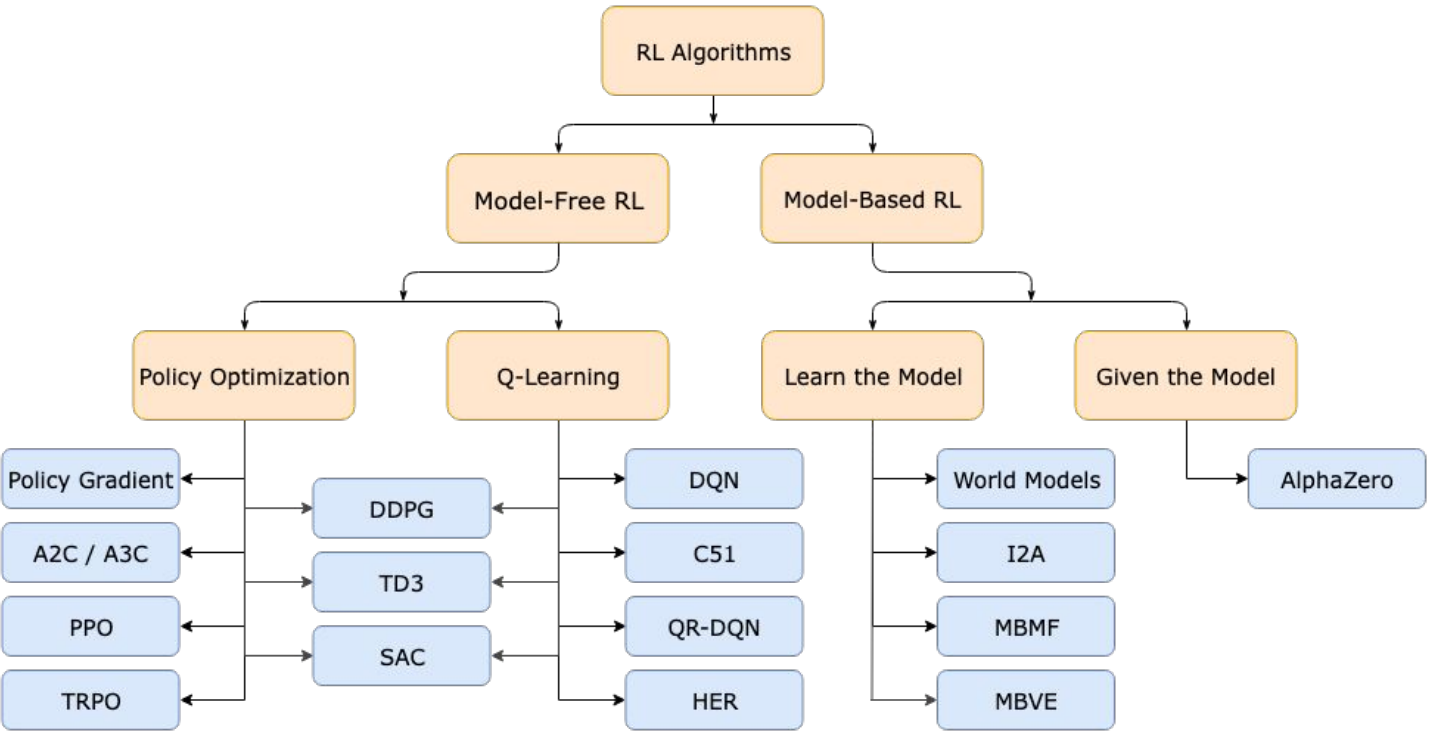
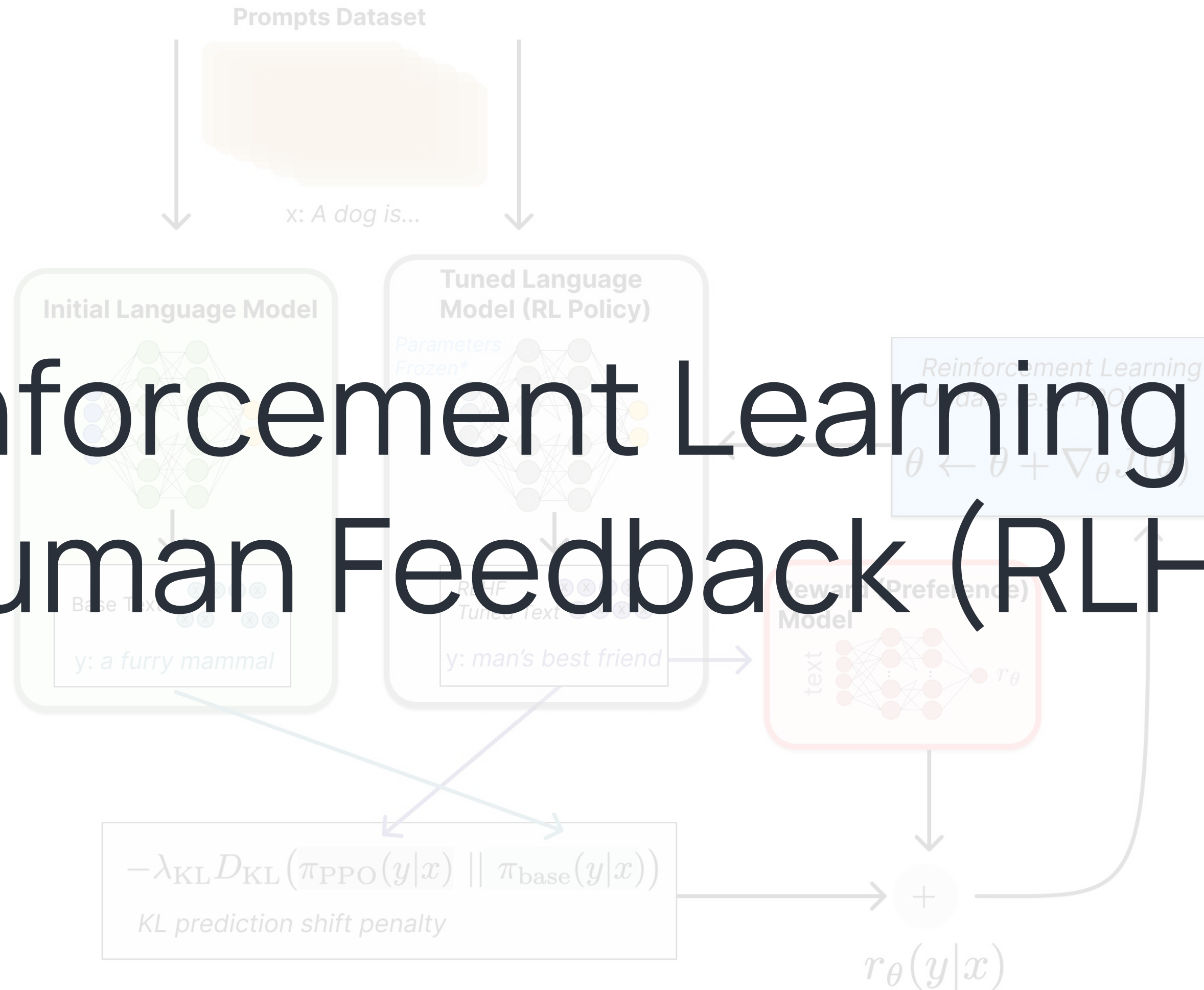


Image credit: BANDAI NAMCO Entertainment America Inc



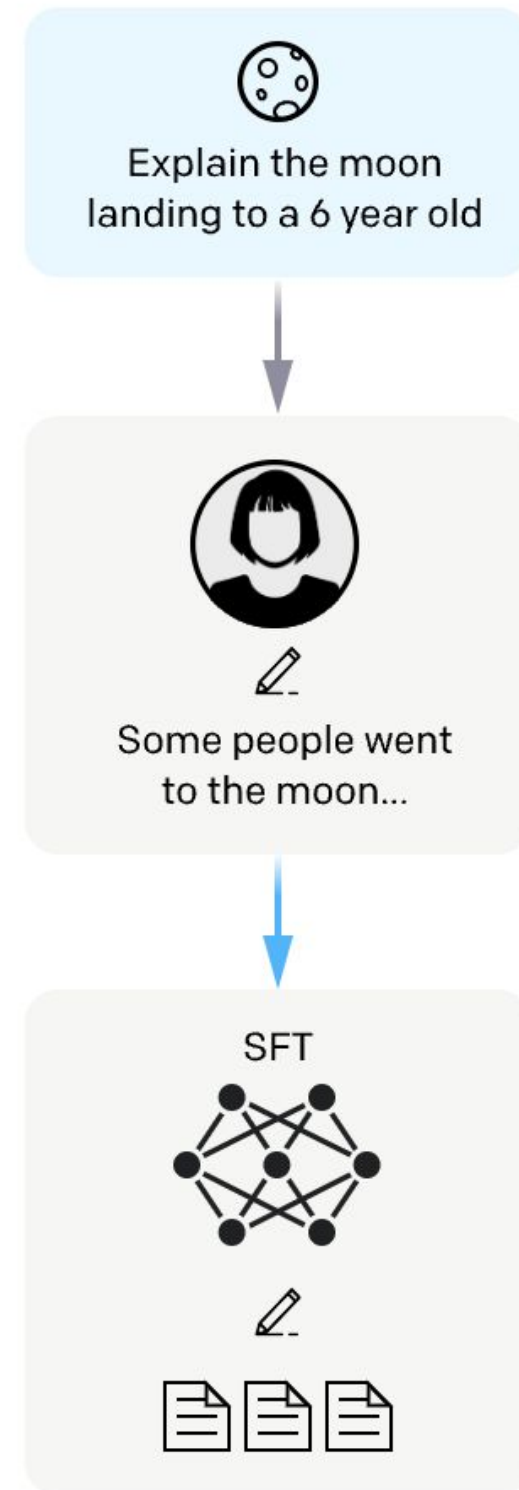
Graphic: OpenAI Spinning Up

Reinforcement Learning from Human Feedback (RLHF)



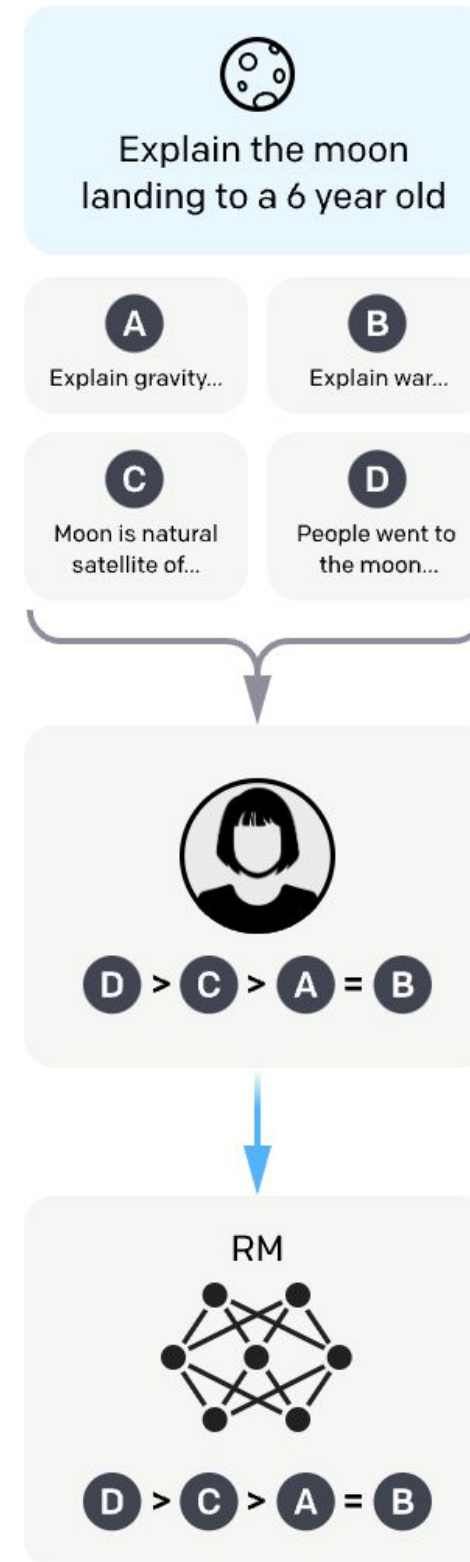
Step 1

**Collect demonstration data,
and train a supervised policy.**



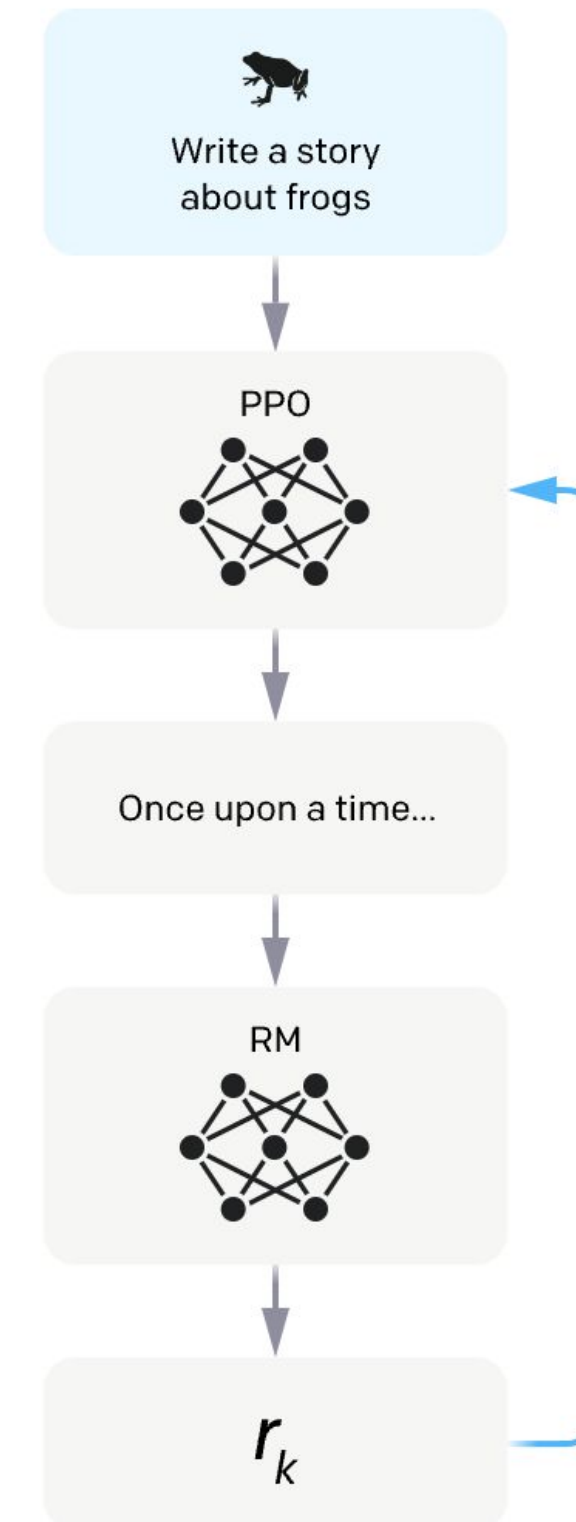
Step 2

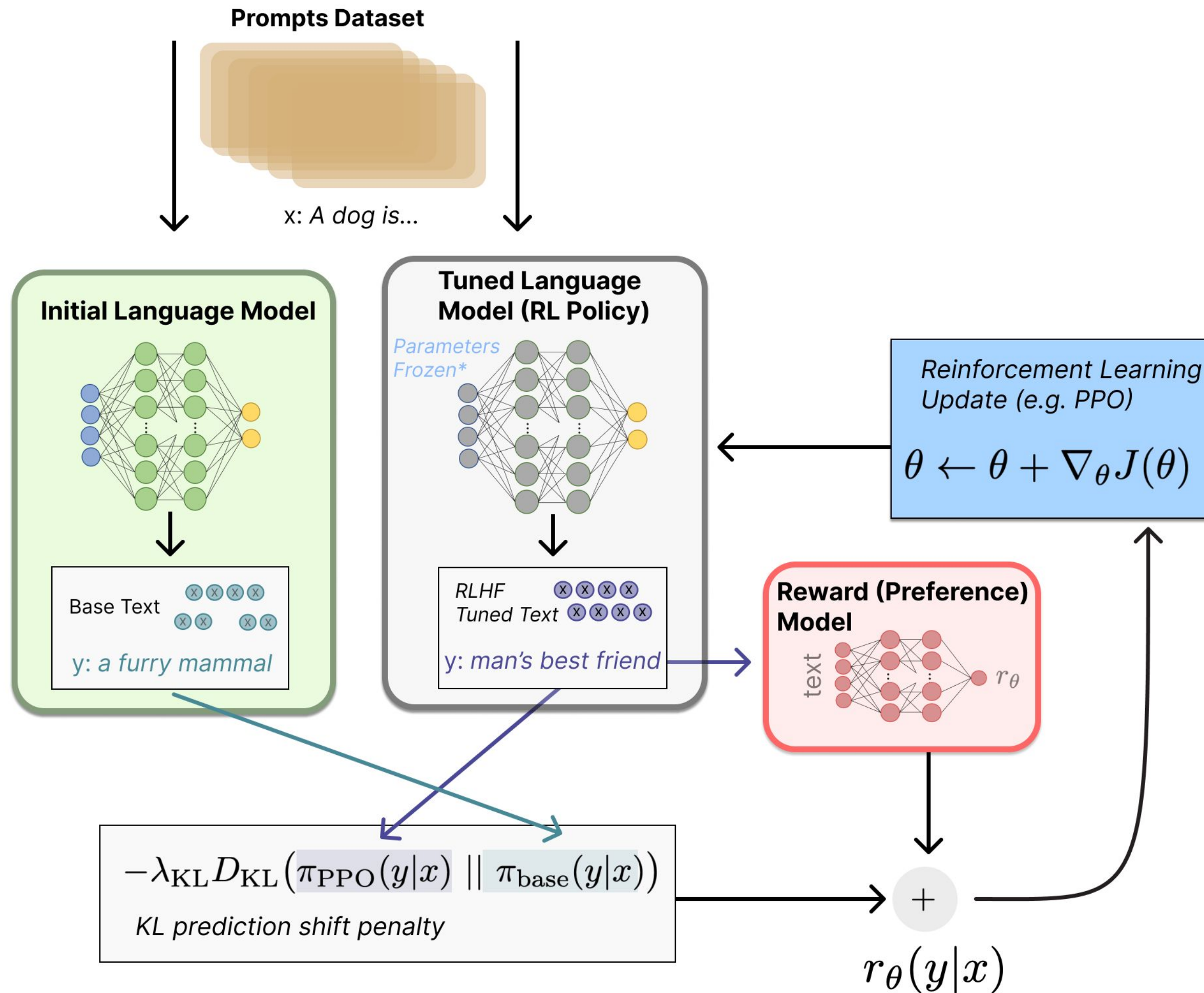
**Collect comparison data,
and train a reward model.**

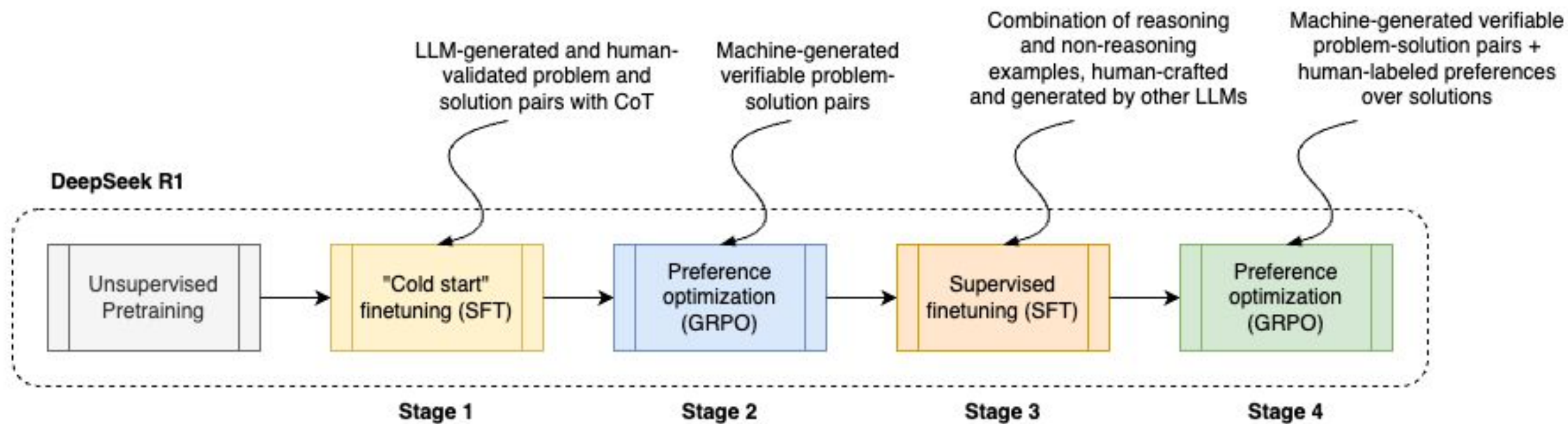
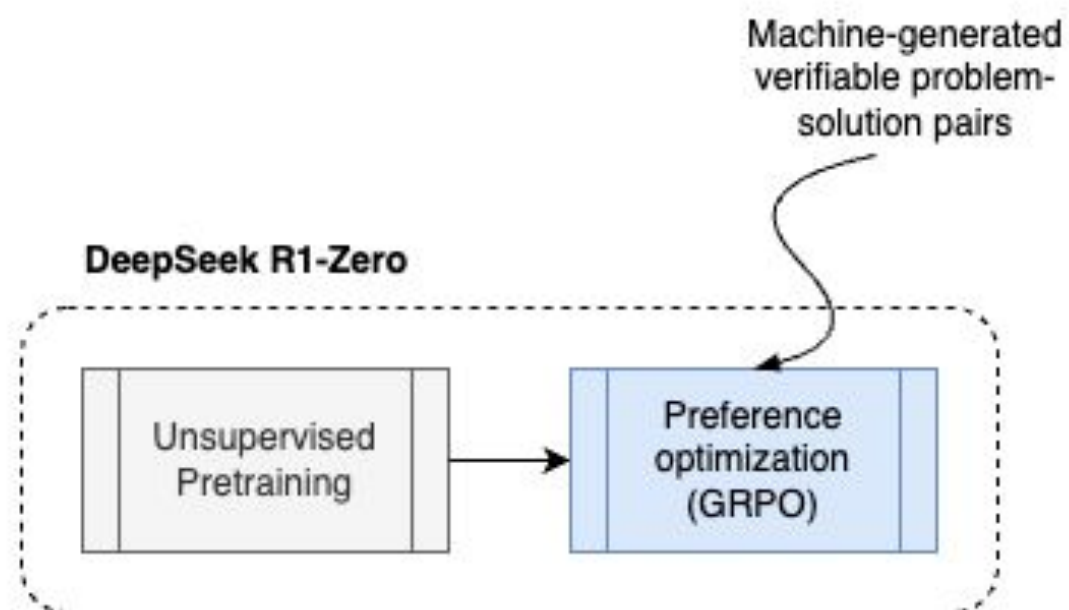
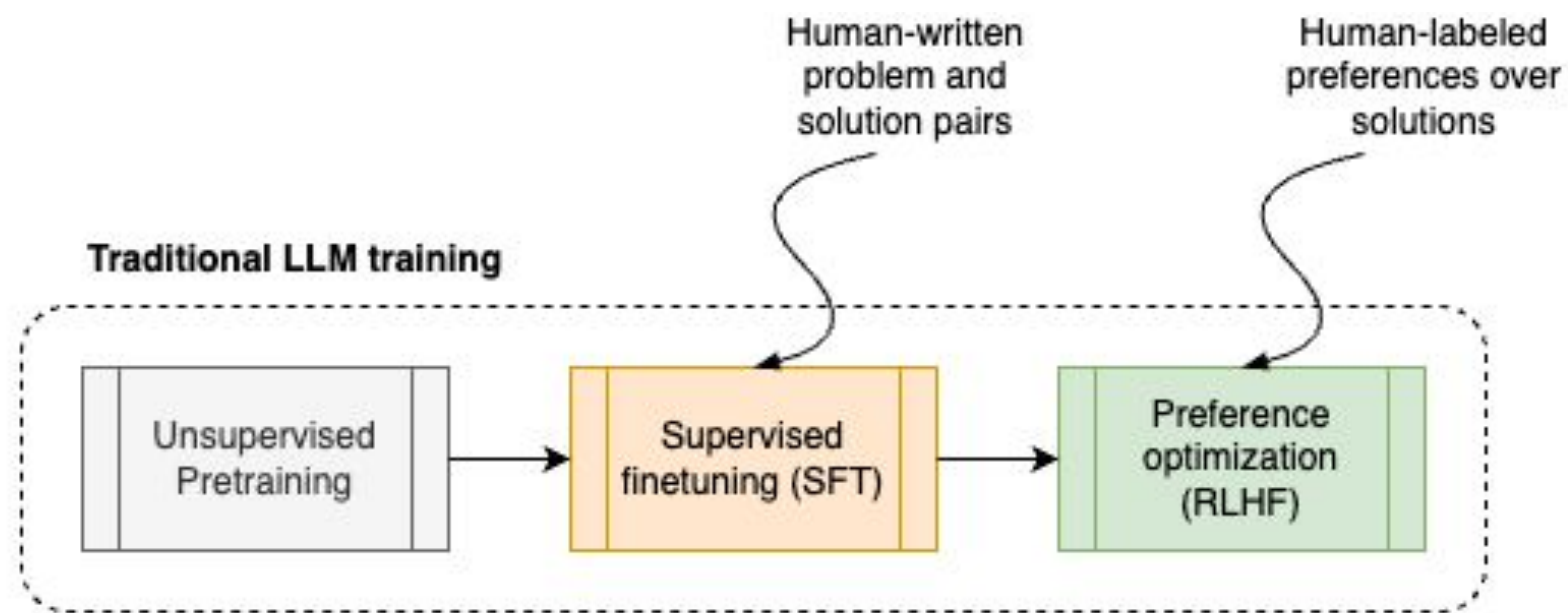


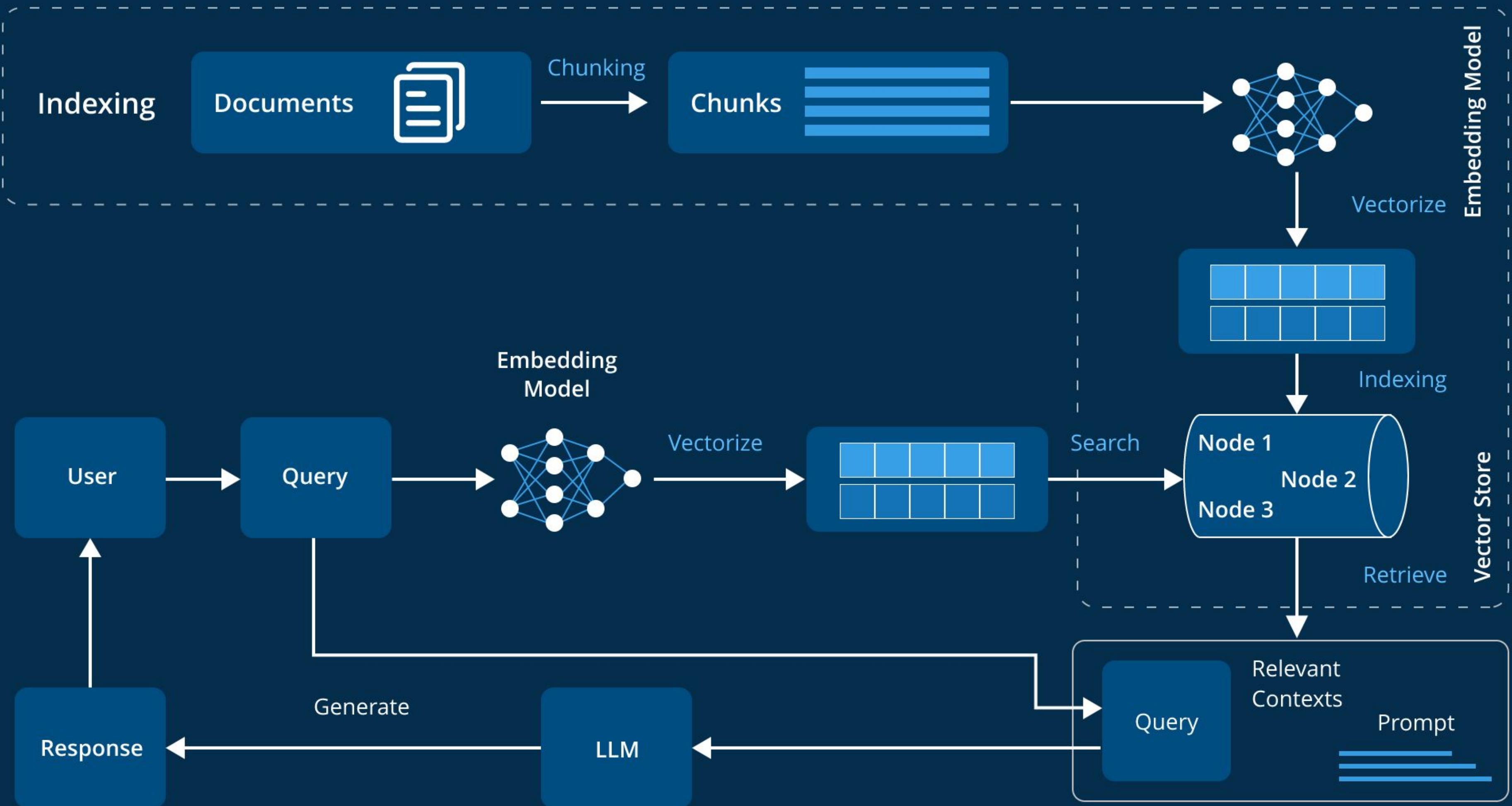
Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**









Demo

Explore Embeddings Live

Goals:

- Gain a deeper understanding of a RAG pipeline through interactive visualizations

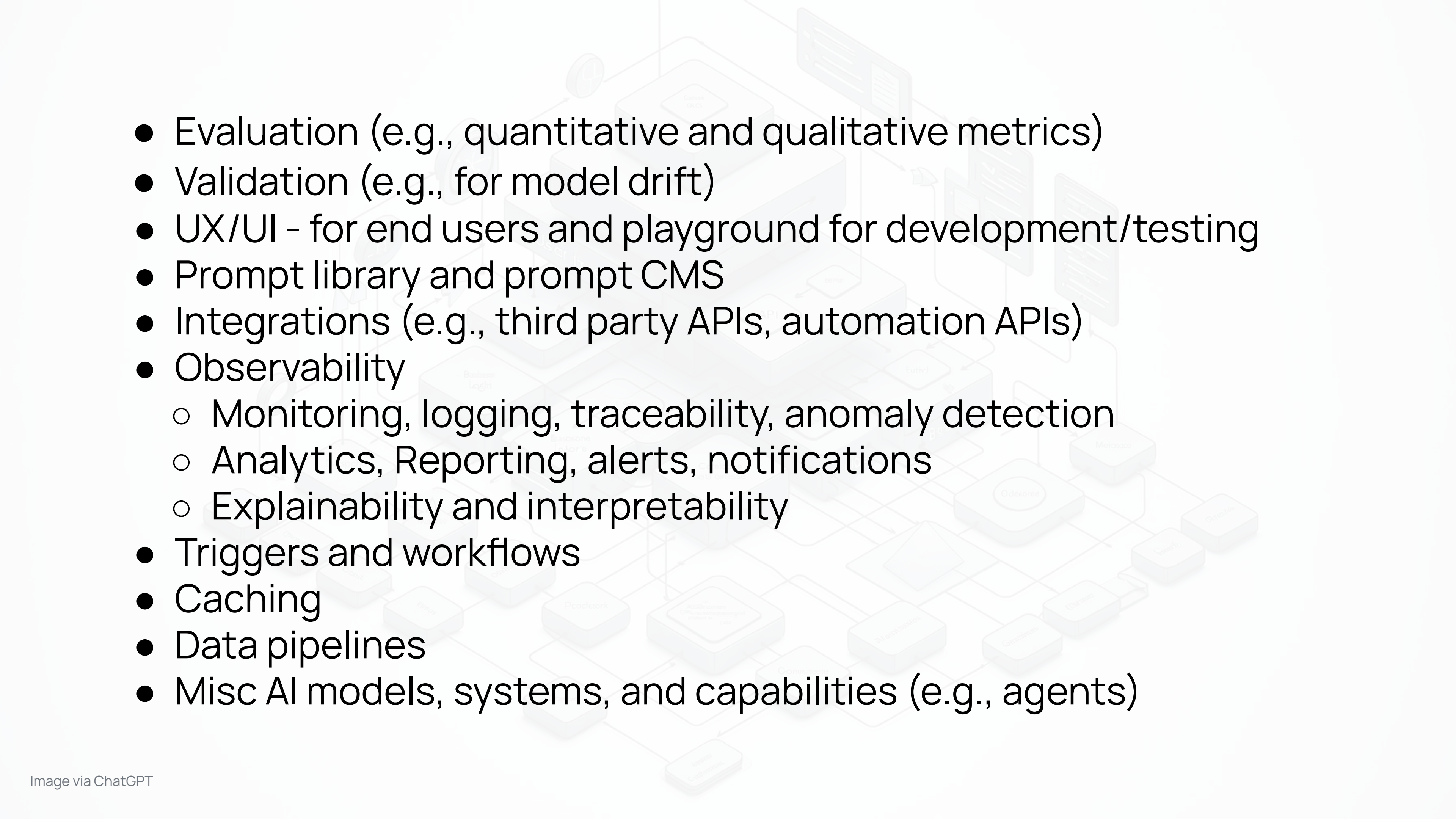
Links:

- [Interactive RAG Playground](#)

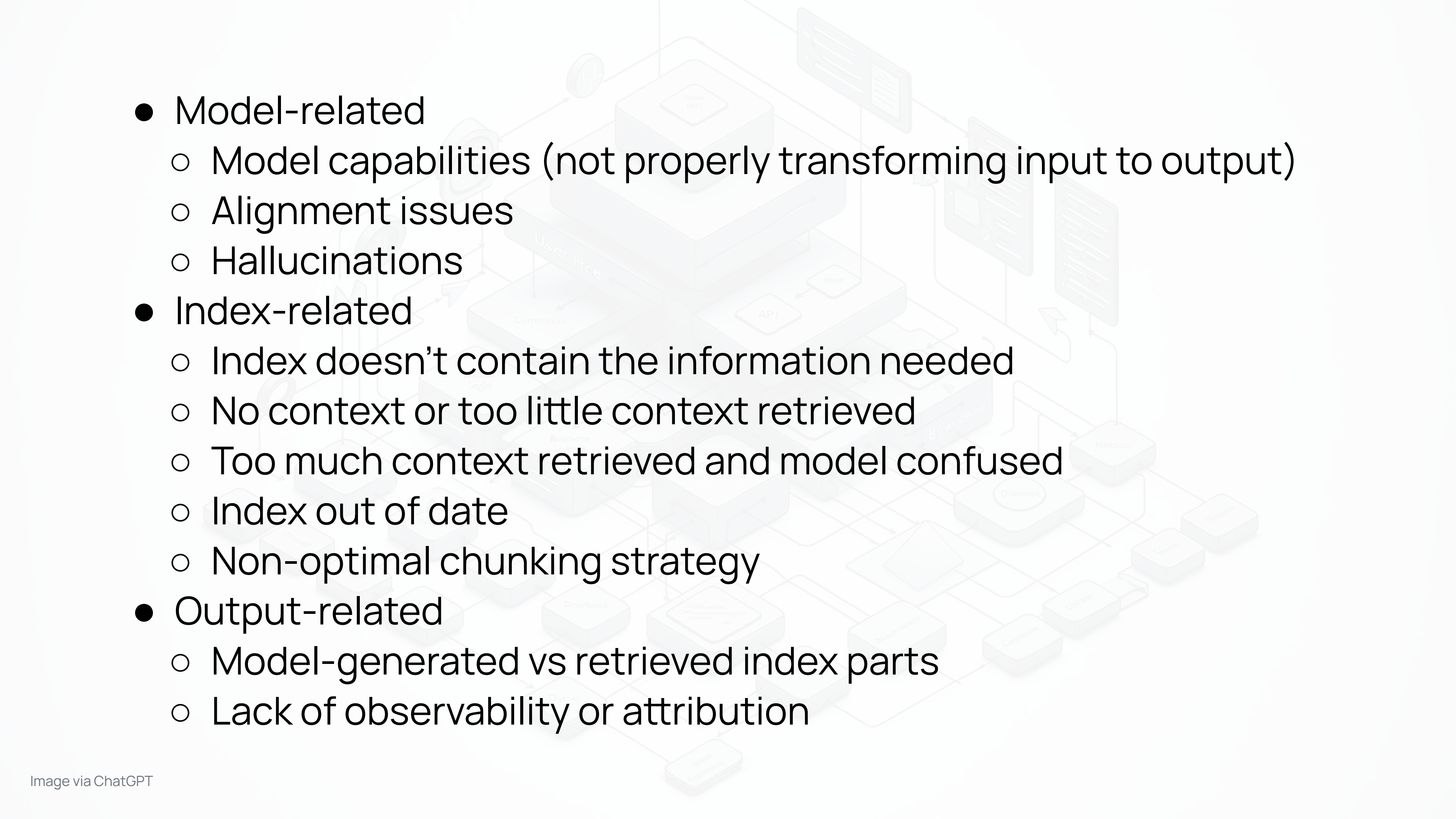
Reflect + Discuss:

- Share your thoughts

What other architectural components might be needed in a production AI system?

- 
- Evaluation (e.g., quantitative and qualitative metrics)
 - Validation (e.g., for model drift)
 - UX/UI - for end users and playground for development/testing
 - Prompt library and prompt CMS
 - Integrations (e.g., third party APIs, automation APIs)
 - Observability
 - Monitoring, logging, traceability, anomaly detection
 - Analytics, Reporting, alerts, notifications
 - Explainability and interpretability
 - Triggers and workflows
 - Caching
 - Data pipelines
 - Misc AI models, systems, and capabilities (e.g., agents)

Why might RAG outputs be inaccurate or untrustworthy?

- 
- Model-related
 - Model capabilities (not properly transforming input to output)
 - Alignment issues
 - Hallucinations
 - Index-related
 - Index doesn't contain the information needed
 - No context or too little context retrieved
 - Too much context retrieved and model confused
 - Index out of date
 - Non-optimal chunking strategy
 - Output-related
 - Model-generated vs retrieved index parts
 - Lack of observability or attribution



AI Risk, Responsibility, & Governance

NIST AI RMF Playbook

Safe

Secure &
Resilient

Explainable &
Interpretable

Privacy-
Enhanced

Fair - With Harmful
Bias Managed

Valid & Reliable

Accountable
&
Transparent

AI Risk Management Framework



Appendix

Learn more about RL @

**OpenAI**
Spinning Up

latest

Search docs

USER DOCUMENTATION
Introduction
Installation
Algorithms
Running Experiments
Experiment Outputs
Plotting Results

INTRODUCTION TO RL
Part 1: Key Concepts in RL
Part 2: Kinds of RL Algorithms
Part 3: Intro to Policy Optimization

RESOURCES
Spinning Up as a Deep RL Researcher
Key Papers in Deep RL
Exercises
Benchmarks for Spinning Up Implementations

ALGORITHMS DOCS
Vanilla Policy Gradient
Trust Region Policy Optimization
Proximal Policy Optimization
Deep Deterministic Policy Gradient
Twin Delayed DDPG
Soft Actor-Critic

UTILITIES DOCS
Logger
Plotter
MPI Tools
Run Utils

ETC.
Acknowledgements

Read the Docs

v: latest

Welcome to Spinning Up in Deep RL!



User Documentation

- [Introduction](#)
 - [What This Is](#)
 - [Why We Built This](#)
 - [How This Serves Our Mission](#)
 - [Code Design Philosophy](#)
 - [Long-Term Support and Support History](#)
- [Installation](#)
 - [Installing Python](#)
 - [Installing OpenMPI](#)
 - [Installing Spinning Up](#)
 - [Check Your Install](#)
 - [Installing MuJoCo \(Optional\)](#)
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 - [Algorithm Outputs](#)
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 - [Loading and Running Trained Policies](#)
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Introduction to RL